Exam 03 Review — Modules 12–16

BIOL-1

Name: _____ **Date:** _____

Learning Objectives

By the end of this review lab, you will be able to:

1. Connect the major ideas from Modules 12–16 across evolution, speciation, ecology, and systems synthesis.
2. Explain key terms in your own words before checking notes or answer keys.
3. Practice short-answer reasoning for the third unit without relying on memorized wording.
4. Identify which concepts need more review before Exam 03.

Purpose

Use this worksheet to connect **evolution**, **speciation**, **ecology**, and **capstone systems synthesis**. Explain ideas in your own words, compare evidence, and identify what you still need to practice.

How to use this lab

1. Do the **written** items **closed-book** first, then reopen notes only where you were stuck. A short **check** (A–D) list at the end of each module is optional—use it to verify you still recognize exam-style phrasing.
2. For **short answers**, write in full sentences (or labeled bullet lists where that matches your instructor's expectations). **Definitions** should use **your** words, not a textbook sentence—then compare to the key terms list.
3. Ask your instructor for the **official exam** instructions, timing, and point breakdown—those rules override anything informal here.

Module 12 — Darwin and evolution

Focus ideas

- **Descent with modification** (change over time from shared ancestors).
- **Natural selection:** heritable **variation**, **overproduction** of offspring, **differential reproductive success** (fitness = relative reproductive success).
- **Individuals** do not evolve; **populations** change in allele frequency over generations.
- **Evidence** from fossils, biogeography, comparative anatomy (homology), molecular comparisons.

Written practice (free response)

1. In **one** clear sentence, define **evolutionary fitness** the way a biologist uses the term in this unit (not "strongest" or "survives longest" unless you explain why that could still matter to fitness).

2. In **2–3** sentences, explain **"descent with modification"** so a friend in another class would understand: ancestry, **change over time**, and why **not** all descendants look identical.

3. Scenario: A tall plant and a short plant in the **same** population both survive a drought **equally well**, but the tall one produces more seeds. In **2–3** sentences, identify the difference in **reproductive success**, state what inheritance evidence you need, and propose an environmental explanation to test. Explain why equal survival and different seed counts alone do not identify the selective pressure.

4. Name **one** line of **evidence** for evolution (fossil, comparative anatomy, biogeography, or molecular) and state **what pattern** that evidence shows—**two** sentences max.

5. In **2–3** sentences, explain why "**populations** evolve, not individuals" using **allele frequencies** (or gene-pool change), not the word *fitness* alone.

6. Optional contrast: Acclimation (one body over time) vs **evolution** (across generations). **One** sentence each.

Quick check — write the letter (A–D) *optional*

A. In biology, **evolutionary fitness** is best defined as:
A) Physical strength relative to other species
B) Survival time without food
C) Relative reproductive success (passing alleles to the next generation)
D) Ability to acclimate to cold weather in one lifetime

B. Which is **strong** evidence of **common descent**?
A) Two species on different continents, so they must be unrelated
B) The same **homologous** bone layout in forelimbs of a whale, a bat, and a human, with different functions
C) Unrelated lineages that look alike because of convergent selection
D) A single individual that lived 200 years and changed its traits

Module 13 — How populations evolve (microevolution)

Focus ideas

- **Microevolution** = **change in allele frequencies** in a population over time.
- **Allele-frequency processes:** mutation, **gene flow**, **genetic drift**, and **natural selection**. **Nonrandom mating** changes genotype proportions and can be associated with sexual selection; alone it need not change allele frequencies.
- **Drift** matters most in **small** populations; **bottleneck** vs **founder** effects.
- **Selection modes** on a continuous trait: **directional**, **stabilizing**, **disruptive**.
- **Hardy–Weinberg** predicts genotype proportions under explicit assumptions. A mismatch does not identify a cause or prove allele-frequency change by itself.

Written practice (free response)

1. Mechanisms comparison: Explain how **mutation**, **gene flow**, **genetic drift**, and **natural selection** can change allele frequencies. Then explain why **nonrandom mating** can change genotype proportions without necessarily changing allele frequencies.

2. Drift vs selection: A hurricane **randomly** kills 90% of a moth population; survivors repopulate. In **2–3** sentences, say whether this is best viewed as **selection** or **drift** and name the specific **drift** label if it applies. Then give a **separate** two-sentence example where **differential survival of genotypes** is **selection**, not random sampling.

3. Founder vs bottleneck: A few lizards **found** a new island population; a storm **bottlenecks** a large mainland population. In **2–3** sentences, contrast what each event does to **allele variety** in the long run (same? different? why?).

4. Selection shapes: For **each** of **directional**, **stabilizing**, and **disruptive** selection, give **one** short phrase: what the environment favors (e.g. “favors the middle of a birth-weight range”).

5. Hardy–Weinberg: In **2–3** sentences, say what the **equilibrium** is **used for** in this course (a word like **null** or **baseline** helps) and name **two** real-world violations that would break it (not a full list—**two** is enough).

6. Two terms you confuse (your words):

Quick check — write the letter (A–D) *optional*

A. Which option is a **mechanism of microevolution**?
A) A sunburn that is not passed to children
B) **Gene flow** when individuals enter the population and mate
C) A population count going up because of immigration without allele change
D) A species inventing a new organ when needed

B. A few lizards are accidentally transported to a new island; the new population only reflects the alleles the founders carried. This is:
A) The **founder** effect (genetic drift)
B) The bottleneck effect in the source population on the mainland
C) Convergent evolution
D) Interspecific competition

Module 14 — Macroevolution and speciation

Focus ideas

- **Biological species concept** (interbreed in nature → viable fertile offspring; **reproductive isolation** from other such groups).
- **Prezygotic** barriers (before zygote): habitat, **temporal**, behavioral, mechanical, gametic, etc.
- **Postzygotic** barriers (after zygote): hybrid viability, hybrid fertility, hybrid breakdown.
- **Allopatric** speciation: geographic split then divergence. **Sympatric** speciation: same place (e.g. **polyploidy** in plants).

Written practice (free response)

1. Prezygotic vs postzygotic: For **(a)** different spawning months, **(b)** hybrid offspring sterile, and **(c)** different courtship songs, label each as **prezygotic** or **postzygotic** and give a **one-word** subtype where you can (temporal, behavioral, etc.).

2. Horse × donkey → mule: In **2–3** sentences, explain which kind of **reproductive barrier** the **sterile mule** represents and **why** that still **blocks** gene flow between the parent species over time.

3. Allopatric pattern: A mountain range splits one species into two populations that no longer **exchange** genes. In **2–3** sentences, describe the **allopatric** story (geography + time + divergence) and one **kind** of difference that might accumulate in each lineage.

4. Sympatric in plants: Explain **sympatric** speciation by **polyploidy** in **2–3** sentences: same place, new chromosome set, and why isolation can be **sudden** compared with slow geographic split.

5. Macroevolution definition: In **2–3** sentences, explain why "macroevolution" in this course is **not** a separate *kind* of DNA magic from "microevolution" but **larger-scale** outcomes built from the same population-level processes.

6. Reproductive isolation and divergence: In **2–3** sentences, explain why, once two lineages **do not** swap genes, they can **accumulate different** adaptations in different environments (use **allele** or **trait** change language).

Quick check — write the letter (A–D) *optional*

A. A horse and a donkey can produce a mule, but the mule is typically sterile. This outcome is:
A) A prezygotic gametic failure
B) A **postzygotic** barrier to gene flow (reduced hybrid fertility)
C) Stabilizing selection in horses
D) The founder effect

B. A mountain range forms and cuts a **single** species into two populations that no longer exchange genes. Over time they diverge until they do not interbreed. This pattern fits:

A) **Allopatric** speciation
B) Only sympatric speciation with polyploidy
C) Macroevolution without microevolution
D) A species concept based only on appearance

Module 15 — Population and systems ecology

Focus ideas

- **Exponential (J)** vs **logistic (S)** growth; **carrying capacity (K)**.
- **Density-dependent** (competition, disease, predation often intensify with crowding) vs **density-independent** (e.g. many fires, some storms) regulation.
- **Trophic levels** and **~10% energy** transfer; **energy flows**, **nutrients** cycle.
- **Producers, consumers, decomposers**; food chains limited in length partly because of energy loss as **heat** and work.

Written practice (free response)

1. J vs S and K: In **3–4** sentences, describe how a **logistic** curve **differs** from unrestrained **exponential** growth, and what **K** means for a real population (you may use a short example: elk, plants, or bacteria).

2. Density (two examples): Give **one** example of a **density-dependent** limit and **one** of a **density-independent** disturbance, each in a **full sentence** with **why** the label fits.

3. Energy vs nutrients: Contrast **energy** flow and **nutrient** cycling in an ecosystem in **3–4** sentences, including decomposers and **why** sunlight is always the **new** energy input.

4. Ten percent in words: With **~10%** transfer per trophic level, explain in **2–3** sentences why supporting **the same** number of people with **tuna** (long chain) generally requires **more** primary production than eating **closer to plants**—clear logic, no need for exact numbers if your reasoning is sound.

5. Connect *K* and energy (one sentence, your words): Link **carrying capacity** to **trophic** levels—why a big base of **producers** supports **few** top consumers.

Quick check — write the letter (A–D) *optional*

A. A disease spreads more easily when hosts are very crowded. That limiting factor is best classified as:
A) Density-**independent**
B) Density-**dependent**
C) Only abiotic (never biological)
D) Not part of population ecology

B. In ecosystems, **chemical elements** (e.g. nitrogen, carbon in nutrients) are **cycled**, while **energy** is:
A) Cycled the same way as nitrogen
B) **Lost as heat** along the way; not recycled to be photosynthesized again from dead matter in the same way sunlight is new input
C) Only stored in apex predators
D) Only found in the atmosphere, never in plants

Word bank sprint

Use each term once. Practice meaning in context, not one memorized sentence shape.

Word bank: *allele frequencies, allopatric, biological species, density-dependent, descent with modification, decomposers, directional selection, homologous, prezygotic, stabilizing selection*

1. In formal terms, **microevolution** means a change in _____ from one generation to the next in a population.

2. In comparative anatomy, forelimb bones in different tetrapods that share structure from a common ancestor are called _____ structures (even if functions differ).

3. A population's birth-weight curve can favor the middle; this mode of natural selection is _____ selection.

4. A barrier that works **before** a zygote forms—such as different breeding seasons—is a _____ barrier to reproduction.

5. A river splits a range; populations diverge in isolation—speciation that begins with **geography** is often _____ speciation.

6. Darwin's phrase for evolution: species share ancestry and accrue change over time—_____ with modification.

7. A limiting factor that hits harder as a population **crowds** (e.g. more transmission of a parasite) is _____ regulation in many textbook examples.

8. A group of populations that interbreed in nature, producing viable fertile offspring, and is reproductively isolated from other such groups, fits the _____ species concept (as presented in this unit).

9. Fungi and bacteria that recycle dead organic matter and return elements to the soil for producers are _____.

10. When a population's average trait value shifts toward **one** extreme of the range (e.g. darker moths on sooty trees), the pattern is _____ selection.

Module 16 — Capstone systems synthesis

Connect a **cause**, a **feedback or constraint**, and **evidence** across biological scales.

1. A drought-tolerant plant variant becomes more common while herbivores decline. Propose a chain connecting the variant, plant reproduction, and herbivore resources. Which links are hypotheses rather than supplied observations?

2. Name one balancing feedback or resource constraint that could limit your predicted change. Explain its direction.

3. Give an alternative explanation for the herbivore decline. What measurement or comparison could distinguish the two explanations? What result would make you revise your claim?

Short-answer rehearsal

Draft at least three in your own words. On your exam, you may answer a subset of longer prompts; check your real instructions.

1. (Module 12) Contrast how **individuals** can change in a lifetime with how **evolution** is defined in this course. Name **one** type of **evidence** for evolution in one sentence.

2. (Module 13) Compare **genetic drift** to **natural selection** (what drives change in each?). Give **one** example of **bottleneck** or **founder** effect in plain words.

3. (Module 14) Classify these as **prezygotic** or **postzygotic**: (a) different courtship song; (b) hybrid is sterile. Then state one difference between **allopatric** and **sympatric** speciation in one or two sentences.

4. (Module 15) Draw the distinction between **density-dependent** and **density-independent** limits, then use the **~10% rule** to explain why long food chains support **fewer** high-level consumers than a base of the same size could support if humans ate **primary producers** directly. (Equations optional; clear logic is enough.)

5. (Synthesis, Module 12–13) A small island population of birds loses genetic diversity after a storm. In **2–3** sentences, explain **one** way this could be **drift** and **one** way a biologist would **not** call the same event "selection."

6. (Synthesis, Module 14–15) A **physical barrier** (new canyon) splits a fish population, and on one side, **prey** are scarce. In **2–3** sentences, tie **allopatric** divergence to how **K** or **trophic** pressure could **differ** between the two sides over time (conceptual, not a math problem).

Self-check before the exam

I can…	✓
State natural selection as variation + overproduction + unequal contribution to the next generation	[ ]
Give **one** type of **evidence** (fossil, anatomy, biogeography, molecular) and what it supports	[ ]
Compare allele-frequency processes and nonrandom mating; contrast drift vs **gene flow**	[ ]
Describe directional / stabilizing / **disruptive** selection; explain bottleneck **vs** founder	[ ]
Classify prezygotic **vs** postzygotic; allopatric **vs** sympatric; biological species idea	[ ]
Contrast J vs S growth; K; density-dependent **vs** independent; energy **flow vs** nutrient **cycling**; 10% rule	[ ]

| Build a cross-scale mechanism chain, identify a feedback or constraint, and propose evidence that could change my explanation | [ ] |

Two terms I still mix up:

One action before the exam (be specific):

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